



NATIONAL INSTITUTE OF TECHNOLOGY,
KURUKSHETRA-136119

B. Tech. (PIE) – 4th Sem.; Mid-SEM Exam – I
February - 2026

PIPC-209

Time: 50 Minutes

Production Technology - II

Max. Marks: 15

Note: Attempt all questions. Assume suitable missing data, if any.

Q-1: Explain the thread grinding process with its principle, construction, and working. Classify the different types of thread grinding methods, and compare it with other thread manufacturing processes in terms of accuracy, surface finish, production rate, cost, and applications. (4)

Q-2: Give suitable shear angle for the punch to bring the work within the capacity of a 45T press, when a hole of 50 mm radius is to be punched in a copper plate of 5.6 mm thickness. The ultimate shear stress is 225 N/mm^2 . The desired cutting is performed at 30% penetration of the punch with normal clearances on the tools. (5)

Q-3: Discuss the relationship between tensile yield stress and shear yield stress developed during metal shearing processes. Further, elaborate the important terminologies involved such as clearance, angular clearance, and punching (or blanking) force. (3)

Q-4: Explain the manufacturing of threads using the casting process, describing its principle and procedure. Classify the suitable casting methods for thread production and discuss pattern considerations and allowances and also state its merits, limitations, defects, and applications. (3)

End of Question Paper